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SUBJECT CODE NO: - RNP-8049
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (CSE/CS/CE/IT) (NEP) Sem - IV
Examination May/June-2026
Operating System

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from question no's 2, 3, 4, 5, 6 and 7

Q.1 Solve any TEN from the following questions**20**

- a) Define multitasking.
- b) What is time sharing system?
- c) Define critical section.
- d) What is mutual exclusion?
- e) What is demand paging?
- f) Define thrashing.
- g) What is file directory?
- h) What is swapping?
- i) Define process hierarchy.
- j) What is I/O controller?
- k) Define deadlock prevention.
- l) What is LRU algorithm?
- m) What is real-time OS?

Q.2 a) Explain evolution of Operating Systems.**5**

- b) Explain OS structures (Monolithic, Layered, Microkernel).

5**Q.3 a) Explain process creation and termination.****5**

- b) Explain Dining Philosopher problem.

5**Q.4 a) Explain file system implementation.****5**

- b) Explain disk space management techniques

5**Q.5 a) Explain logical vs physical address space.****5**

- b) Explain page replacement algorithms (FIFO, LRU).

5**Q.6 a) Explain I/O software layers.****5**

- b) Explain disk formatting and RAID.

5**Q.7 a) Explain deadlock avoidance with Banker's algorithm.****5**

- b) Compare Linux OS and Android OS

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